

SPECTRAL THEORY FOR SELF-ADJOINT OPERATORS

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References

- Spanish
 - Kenneth Hoffman y Ray Kunze. *Algebra Lineal*. Prentice Hall
- English:
 - Kenneth Hoffman and Ray Kunze. *Linear Algebra* . Pearson

Vector spaces and inner products:

Definitions

- Let us consider the N -dimensional complex space $\mathbb{C}^N$
- We define the inner product in $\mathbb{C}^N$:
 - If $u = \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ \vdots \\ v_N \end{bmatrix}$ then $\langle u | v \rangle := \sum u_n \overline{v_n}$
 - Recall that the complex conjugate is defined as $\overline{a + bi} := a - bi$
- Let $\mathcal{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_N\}$ be an orthonormal base in $\mathbb{C}^N$:
 - $\{\mathbf{e}_1, \dots, \mathbf{e}_N\}$ are linearly independent
 - $\langle \mathbf{e}_i | \mathbf{e}_j \rangle = \delta_{ij}$
- Let $T: \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator
 - For a given base, for example $\mathcal{E}$, the associated matrix is defined as $A = [T]_{\mathcal{E}}$
- Let $f: \mathbb{C}^N \rightarrow \mathbb{C}$ be a linear functional
 - Think of linear functionals as projections

Eigenspaces, Eigenvectors and Eigenvalues

- Let $T: \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator
 - $\lambda \in \mathbb{C}$ is an eigenvalue of T if there exist a vector $v \neq 0$ such that $Tv = \lambda v$
 - The vector v above is called an eigenvector (associated to the eigenvalue λ)
 - Let $E(\lambda)$ be the set of all eigenvectors associated to an eigenvalue λ
- **Proposition 1.** $E(\lambda)$ is a linear subspace of $\mathbb{C}^N$
- Proof:
 - We only need to use the linearity of T and the definition of eigenvalue
 - For any $u, v \in E(\lambda)$ and any $a, b \in \mathbb{C}$ we have:
$$T(au + bv) = aT(u) + bT(v) = a(\lambda u) + b(\lambda v) = \lambda(au + bv)$$

Linear functionals as projections:

Uniqueness of the projection

- **Proposition 2.** For any linear functional f there exists a unique vector v_f such that for every $u \in \mathbb{C}^N$ we have $f(u) = \langle u | v_f \rangle$

- Proof:

➤ For the orthonormal base $\mathcal{E} = \{e_1, \dots, e_N\}$ define $v_f = \sum \overline{f(e_n)} e_n$

➤ Define the new functional $f_v(u) = \langle u | v_f \rangle$

➤ Then for any $e_i \in \mathcal{E}$:

$$f_v(e_i) = \langle e_i | v_f \rangle = \langle e_i | \sum \overline{f(e_n)} e_n \rangle = \sum f(e_n) \langle e_i | e_n \rangle = f(e_i)$$

➤ Since f and f_v coincide in all elements of $\mathcal{E}$ then $f = f_v$

➤ If $f(u) = \langle u | w_f \rangle$ then $\langle u | v_f - w_f \rangle = 0$ for all u . Choosing $u = v_f - w_f$ it follows that $\langle u | u \rangle = 0$. Therefore $u = 0$ i.e. $v_f = w_f$

Adjoint or Hermitian operator: Definition and uniqueness

- **Proposition 3.** For any linear operator T there exists a unique linear operator T^* (called the adjoint or Hermitian operator of T) such that for every $u, v \in \mathbb{C}^N$ we have

$$\langle Tu|v\rangle = \langle u|T^*v\rangle$$

- Proof:

- Let $v \in \mathbb{C}^N$ be fixed. Define the linear functional $u \rightarrow \langle Tu|v\rangle$
- From Proposition 2 there is a unique vector $v' \in \mathbb{C}^N$ such that $\langle Tu|v\rangle = \langle u|v'\rangle$
- Define the mapping $T^*: \mathbb{C}^N \rightarrow \mathbb{C}^N$ as $T^*v := v'$
- By definition, $\langle Tu|v\rangle = \langle u|T^*v\rangle$
- For the linearity of T we have:
$$\begin{aligned}\langle u|T^*(av + bw)\rangle &= \langle Tu|av + bw\rangle = \bar{a}\langle Tu|v\rangle + \bar{b}\langle Tu|w\rangle \\ &= \bar{a}\langle u|T^*v\rangle + \bar{b}\langle u|T^*w\rangle = \langle u|aT^*v + bT^*w\rangle\end{aligned}$$
- Hence the uniqueness in Proposition 2 implies $T^*(av + bw) = aT^*v + bT^*w$
- Finally, the uniqueness in Proposition 2 implies the uniqueness of T^*

Matrix representation using orthogonal projections

- **Proposition 4.** Let $\mathcal{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_N\}$ be an orthonormal base in $\mathbb{C}^N$ and let $A = [T]_{\mathcal{E}}$ be the associated matrix. Then $A_{ij} = \langle T\mathbf{e}_j | \mathbf{e}_i \rangle$
- Proof:
 - Any $u \in \mathbb{C}^N$ can be written uniquely as $u = \sum u_i \mathbf{e}_i$
 - Applying the inner product and using the orthogonality in $\mathcal{E}$ we have that for any $\mathbf{e}_i$:
$$u_i = \langle u | \mathbf{e}_i \rangle$$
 - Therefore we can write $u = \sum \langle u | \mathbf{e}_i \rangle \mathbf{e}_i$
 - This is also true for Tu i.e. $Tu = \sum \langle Tu | \mathbf{e}_i \rangle \mathbf{e}_i$
 - Using $u = \mathbf{e}_j$ we obtain $T\mathbf{e}_j = \sum \langle T\mathbf{e}_j | \mathbf{e}_i \rangle \mathbf{e}_i$
 - On the other hand, the matrix A is defined on the elements of $\mathcal{E}$ i.e. $T\mathbf{e}_j = \sum A_{ij} \mathbf{e}_i$
 - The uniqueness of the representation on the base $\mathcal{E}$ yields $A_{ij} = \langle T\mathbf{e}_j | \mathbf{e}_i \rangle$

The Hermitian matrix is the conjugate and transpose

- **Proposition 5.** Let $\mathcal{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_N\}$ be an orthonormal base in $\mathbb{C}^N$. Let $A = [T]_{\mathcal{E}}$ and $B = [T^*]_{\mathcal{E}}$. Then :
 - The Hermitian or adjoint matrix B is such that $B_{ij} = \overline{A_{ji}}$
 - In the case of $\mathbb{R}^N$ the adjoint matrix is the transpose
- Proof:
 - By Proposition 4 we have $A_{ij} = \langle T\mathbf{e}_j | \mathbf{e}_i \rangle$ and $B_{ij} = \langle T^*\mathbf{e}_j | \mathbf{e}_i \rangle$
 - Therefore:
$$B_{ij} = \langle T^*\mathbf{e}_j | \mathbf{e}_i \rangle = \overline{\langle \mathbf{e}_i | T^*\mathbf{e}_j \rangle} = \overline{\langle T\mathbf{e}_i | \mathbf{e}_j \rangle} = \overline{A_{ji}}$$
- We say that an operator T is self-adjoint operator if $T = T^*$
 - Observe that if a matrix is real and symmetric then it is self-adjoint

Self-adjoint operators: real eigenvalues, orthogonal eigenvectors

- **Proposition 6.** If $T: \mathbb{C}^N \rightarrow \mathbb{C}^N$ is a self-adjoint operator then:
 - a) All eigenvalues of T are real
 - b) Eigenvectors from different eigenvalues are orthogonal
- Proof:
 - Let $\lambda \in \mathbb{C}$ and $v \neq 0$ be such that $Tv = \lambda v$. Then:
$$\lambda \langle v|v \rangle = \langle \lambda v|v \rangle = \langle Tv|v \rangle = \langle v|Tv \rangle = \langle v|\lambda v \rangle = \bar{\lambda} \langle v|v \rangle$$
 - In consequence $\lambda = \bar{\lambda}$
 - Let (λ_i, v_i) and (λ_j, v_j) be eigenvalues and eigenvectors with $\lambda_i \neq \lambda_j$. Then:
$$\lambda_i \langle v_i|v_j \rangle = \langle \lambda_i v_i|v_j \rangle = \langle Tv_i|v_j \rangle = \langle v_i|Tv_j \rangle = \langle v_i|\lambda_j v_j \rangle = \lambda_j \langle v_i|v_j \rangle$$
 - In consequence $\langle v_i|v_j \rangle = 0$

Eigenvalues, eigenvectors and determinants

- **Proposition 7.** Let $A: E \rightarrow E$ where $E = \mathbb{R}^N$ or $E = \mathbb{C}^N$. The following statements are equivalent:
 - a) λ is an eigenvalue of A
 - b) The operator $(A - \lambda I)$ is singular i.e. non-invertible
 - c) $\det(A - \lambda I) = 0$
- Proof:
 - If λ is an eigenvalue of A then there exist $v \in E \setminus \{0\}$ such that $Av = \lambda v$
 - But that means that $(A - \lambda I)v = 0$, hence $(A - \lambda I)$ has a non-trivial kernel
 - In finite dimension, $(A - \lambda I)$ is non-invertible iff $\det(A - \lambda I) = 0$
 - If we assume $\det(A - \lambda I) = 0$ then there is a vector $v \in E \setminus \{0\}$ in $\text{Ker}(A - \lambda I)$ i.e. λ is an eigenvalue of A

Self-adjoint operators and symmetric matrices

- **Proposition 8.** If $T: \mathbb{C}^N \rightarrow \mathbb{C}^N$ is a self-adjoint operator then:
 - a) T has at least one eigenvector i.e. there exists $\lambda \in \mathbb{R}$ and $v \in \mathbb{C}^N \setminus \{0\}$ such that $Tv = \lambda v$
 - b) If $A = [T]_{\mathcal{E}}$ is a real, symmetric matrix then v can be chosen such that $v \in \mathbb{R}^N$
- Proof:
 - By Proposition 7, $Tv = \lambda v$ iff $\det(T - \lambda I) = 0$
 - Since $\det(T - \lambda I)$ is a polynomial in $\mathbb{C}$ then it has a root $\lambda \in \mathbb{C}$
 - In consequence, there exists $v \in \mathbb{C}^N \setminus \{0\}$ such that $Tv = \lambda v$
 - Moreover, by Proposition 6 we have that $\lambda \in \mathbb{R}$
 - If A is real and symmetric then $\det(A - \lambda I)$ is a polynomial in $\mathbb{R}$ and has a root $\lambda \in \mathbb{R}$
 - By Proposition 7, there exists $v \in \mathbb{R}^N \setminus \{0\}$ such that $Tv = \lambda v$

Orthogonal spaces and projections

- Let U be a linear subspace of $\mathbb{C}^N$ and let $\{\mathbf{e}_1, \dots, \mathbf{e}_K\}$ be an orthonormal base of U

➤ The operator $P: \mathbb{C}^N \rightarrow U$ defined as

$$P(x) = \sum_{n=1}^K \langle x | \mathbf{e}_n \rangle \mathbf{e}_n$$

is called the orthogonal projection of $\mathbb{C}^N$ onto U

➤ We denote the orthogonal complement of U in $\mathbb{C}^N$ as

$$U^\perp = \{v \in \mathbb{C}^N : \langle u | v \rangle = 0 \ \forall u \in U\}$$

- Proposition 9.** $U^\perp$ is a linear subspace of $\mathbb{C}^N$ and $\mathbb{C}^N = U \oplus U^\perp$

- Proof:

➤ $0 \in U^\perp$ since $\langle u | 0 \rangle = 0$ for all $u \in U$

➤ For any $v, w \in U^\perp$ and any $a, b \in \mathbb{C}$ we have

$$\langle u | av + bw \rangle = \bar{a} \langle u | v \rangle + \bar{b} \langle u | w \rangle = \bar{a}0 + \bar{b}0 = 0$$

➤ If $u \in U \cap U^\perp$ then $\langle u | u \rangle = 0$, therefore $u = 0$

➤ Let $\{\mathbf{e}_1, \dots, \mathbf{e}_K, \mathbf{e}_{K+1}, \dots, \mathbf{e}_N\}$ be an orthonormal base of $\mathbb{C}^N$ and let $x \in \mathbb{C}^N$. Then:

$$x = \sum_{n=1}^K \langle x | \mathbf{e}_n \rangle \mathbf{e}_n \oplus \sum_{n=K+1}^N \langle x | \mathbf{e}_n \rangle \mathbf{e}_n \in U \oplus U^\perp$$

Invariant spaces for self-adjoint operators

- **Proposition 10.** Let U be a linear subspace and $U^\perp$ its orthogonal complement in $\mathbb{C}^N$. If $T: \mathbb{C}^N \rightarrow \mathbb{C}^N$ is a self-adjoint operator then:
 - a) If U is invariant under T then $U^\perp$ is invariant under T
 - b) In particular, the orthogonal complement of an eigenspace is invariant under T
- Proof:
 - Let $u \in U$. Since U is invariant under T then $Tu \in U$
 - For any $v \in U^\perp$, using that T is self-adjoint we have:
$$\langle u | Tv \rangle = \langle Tu | v \rangle = 0$$
 - In consequence, $Tv \in U^\perp$
 - An eigenspace U is invariant under T because if $u \in U$ then $Tu = \lambda u \in U$

Self-adjoint operators have a base of pure eigenvectors

- **Proposition 11.** Let E be a finite-dimensional linear space with inner product i.e. $E = \mathbb{R}^N$ or $E = \mathbb{C}^N$. If $T: E \rightarrow E$ is a self-adjoint operator then there exists an orthogonal base in E where all elements are eigenvectors of T .
- Proof:
 - By Proposition 8, there exists $\lambda_1 \in \mathbb{R}$ and $v_1 \in E \setminus \{0\}$ such that $Tv_1 = \lambda_1 v_1$
 - Let E_1 be the 1D subspace generated by v_1
 - By Proposition 9 we have $E = E_1 \oplus E_1^\perp$
 - By Proposition 10 it follows that both E_1 and $E_1^\perp$ are invariant under T
 - By induction on the dimension of E , the Proposition holds for $E_1^\perp$ because its dimension is $N - 1 < N$
 - In consequence, there exists an orthogonal base $\{v_2, \dots, v_N\}$ in $E_1^\perp$ where all its elements are eigenvectors of T i.e. $Tv_n = \lambda_n v_n$ for $n = 2, \dots, N$
 - Adding v_1 it follows that $\{v_1, v_2, \dots, v_N\}$ is an orthogonal base in $E = E_1 \oplus E_1^\perp$ where all its elements are eigenvectors of T

Spectral Theorem for self-adjoint operators

- Let us summarise all that we have proven so far in an elegant fashion.
- **Spectral Theorem.** Let E be a finite-dimensional linear space with inner product i.e. $E = \mathbb{R}^N$ or $E = \mathbb{C}^N$. Let $T: E \rightarrow E$ be a self-adjoint operator and let $\lambda_1, \dots, \lambda_K \in \mathbb{R}$ be its distinct eigenvalues. Let E_i be the eigenspace associated to λ_i and let $P_i: E \rightarrow E_i$ be the projection of E to the eigenspace E_i . Then:
 - a) If $i \neq j$ then E_i is orthogonal to E_j
 - b) The space E is the direct sum of all eigenspaces:
$$E = E_1 \oplus \dots \oplus E_K$$
 - c) The operator T admits the following spectral decomposition:
$$T = \lambda_1 P_1 + \dots + \lambda_K P_K$$
- The Spectral Theorem also holds in infinite dimensions (Hilbert spaces and Sobolev spaces), but we need some extra hypotheses on the self-adjoint operator T
 - For example, the Laplacian operator on a compact domain in $\mathbb{R}^N$ with Dirichlet boundary conditions is diagonalizable: this is the basis of Fourier series